

P1a: Magic Madgwick Filter for Attitude Estimation

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Abstract—In this work, attitudes are estimated from a 6-DOF IMU dataset split into six different training movements and one test movement with increasing complexity. The attitudes are estimated with four unique methods: accelerometer-based estimation, gyroscope-based estimation, a complementary filter, and a Madgwick filter. Estimated orientations are then compared to other methods and to a baseline Vicon motion capture system. The following methods are plotted and compared for analysis.

I. INTRODUCTION

This report is intended to serve as a description and analysis of the methods and results used for Project 1a: Magic Madgwick Filter for Attitude Estimation for the class RBE 595-ST: Hands-On Autonomous Aerial Robotics. In this project, an ArduIMU+V2 6-DOF IMU is used to collect six different movements. To estimate the 3D orientation of the IMU and the accuracies of various methods used to estimate the orientation, data collected from the accelerometer and gyroscope are converted to physical units. The Vicon data is also interpolated to align the motion capture system's timestamps with the IMU data. Then four methods of attitude estimation are performed and compared to each other and to the ground-truth Vicon system.

II. DATA PREPARATION

The ArduIMU+V2 produces raw sensor data which must be converted to physical units for valid calculation of orientation. Accelerometer data is converted to m/s^2 and gyroscope data is converted to rad/s .

A. Accelerometer

The data received from the accelerometer is in the format $a = [ax, ay, az]$. The given conversion equation:

$$\tilde{a}_i = \frac{a_i + b_{a,i}}{s_{a,i}} \quad (1)$$

Here $\tilde{a}_i$ is the acceleration measurement in m/s^2 , a_i is the raw measurement, $b_{a,i}$ is the calibrated bias for each axis, and $s_{a,i}$ is the scale factor for each axis. However, this equation produces large acceleration values in the thousand-scale m/s^2 , unlikely to be produced on an IMU in a lab setting. Furthermore, examining the bias values in the IMU Parameters matrix indicated the calculated biases were already in SI units. To resolve this, we assume that the IMU is at rest

and that the accelerometers a_z value should produce a positive $9.81m/s^2$ or g. Thus, the following equation was used instead:

$$\tilde{a}_i = (a_i * s_{a,i} + b_{a,i}) * \mu_i. \quad (2)$$

μ is calculated by dividing the SI measurement of gravity by the average normalized value of the first 100 samples:

$$\mu_i = \sum_{i=0}^{200} \tilde{a}_i \quad (3)$$

B. Gyroscope

The gyroscope from the IMU produces three values $\omega = [\omega_1, \omega_2, \omega_3]$. To convert them to SI units, the following equation was given and used:

$$\tilde{\omega}_i = \frac{3300}{1023} \times \frac{\pi}{180} \times 0.3 \times (\omega_i - b_{i,g}) \quad (4)$$

where, b_i, g is the bias calculated for each gyroscope axis. The average of the first 200 readings of the gyroscope is calculated and used for the above equation.

C. Vicon

The Vicon motion capture system acts as the baseline to compare the attitude estimation equations. However, its sampling frequency and moment of sampling differs from the IMU. Thankfully, each system timestamps their data based on the seconds from a given epoch and both share the same Unix epoch. Therefore, the Vicon motion capture system can simply be interpolated using the python module's `scipy's slerp` function, which takes a list of 3x3 rotations and their corresponding timestamps. The `slerp` function can then produce additional 3x3 rotation matrices based on a new list of timestamps, the IMU's timestamps. `Scipy's slerp` function is based on the Spherical Linear Interpolation which uniformly interpolates between two points on a sphere, enabling interpolation regardless of dimension. This makes it extremely effective for the Vicon capture motion system, although doing so slightly reduces the sample size of each training movement.

III. METHODS

A. Attitude from Gyroscope

The gyroscope provides the body-frame angular velocity $\omega_k = [\omega_{x,k}, \omega_{y,k}, \omega_{z,k}]^T$. The attitude is estimated by numerically integrating these measurements on the rotation group

SO(3). For the interval $\Delta t_k = t_{k+1} - t_k$, the incremental rotation is

$$\Delta \mathbf{R}_k = \text{Exp}([\omega_k]_{\times} \Delta t_k), \quad (5)$$

where

$$[\omega_k]_{\times} = \begin{bmatrix} 0 & -\omega_{z,k} & \omega_{y,k} \\ \omega_{z,k} & 0 & -\omega_{x,k} \\ -\omega_{y,k} & \omega_{x,k} & 0 \end{bmatrix}. \quad (6)$$

Because the angular velocity is expressed in the IMU body frame, the incremental rotation is right-multiplied with the current attitude:

$$\mathbf{R}_{k+1} = \mathbf{R}_k \Delta \mathbf{R}_k. \quad (7)$$

The initial orientation is obtained from the first time-synchronized Vicon measurement,

$$\mathbf{R}_0 = \mathbf{R}_{\text{Vicon}}(t_0). \quad (8)$$

The estimated rotation matrices are then converted to ZYX Euler angles to obtain roll, pitch, and yaw. The gyroscope angle sequences are unwrapped to remove artificial jumps between $+\pi$ and $-\pi$. Although gyro integration accurately captures short-term rotation, residual sensor bias and noise accumulate over time and produce drift from the Vicon ground truth.

B. Attitude from Accelerometer

Assuming that the IMU undergoes rotation with negligible translational acceleration, the calibrated accelerometer vector $\mathbf{a}_k = [a_{x,k}, a_{y,k}, a_{z,k}]^T$ primarily represents the gravity vector projected onto the IMU body axes. The roll and pitch estimates are calculated as

$$\phi_k = \text{atan2}\left(a_{y,k}, \sqrt{a_{x,k}^2 + a_{z,k}^2}\right), \quad (9)$$

$$\theta_k = \text{atan2}\left(-a_{x,k}, \sqrt{a_{y,k}^2 + a_{z,k}^2}\right). \quad (10)$$

The implementation also calculates the following gravity-relative angle:

$$\psi_k^* = \text{atan2}\left(\sqrt{a_{x,k}^2 + a_{y,k}^2}, a_{z,k}\right). \quad (11)$$

Because gravity is unchanged by rotation about the vertical axis, an accelerometer cannot independently determine absolute yaw. Therefore, ψ_k^* is treated as a gravity-relative inclination rather than a true heading measurement.

To reduce high-frequency noise, each accelerometer-derived angle is passed through a first-order low-pass filter:

$$\hat{x}_k = (1 - \alpha)x_k + \alpha\hat{x}_{k-1}, \quad \alpha = 0.8. \quad (12)$$

The accelerometer estimate is most reliable during stationary or low-acceleration motion. During rapid translation, the sensor measures both gravity and linear acceleration, which introduces temporary errors in the estimated attitude.

C. Attitude from Complementary Filter

The gyroscope is accurate over a brief period and quick motions, but it drifts over a long time. The accelerometer doesn't drift and works better over slower motions, but is very noisy in nature. The complimentary filter attempts to fuse them in a way to cover the each other's weaknesses-The gyroscope dominates fast motions while the accelerometer corrects slow drift.

$$\theta_t = \alpha\theta_{gyro,t} + (1 - \alpha)\theta_{accel,t} \quad (13)$$

Eq.13,enforces a high pass filter on the gyroscope estimation and a low pass filter on the accelerometer estimation every time step. A weighted sum is taken between the gyro and accelerometer estimates, where the gyro estimate itself is propagated forward from the previous iteration's fused state. The coefficient α is chosen to be 0.98 in our case. This heavy relies on the gyro while the accelerometer occasionally corrects for drift.

D. Attitude from Madgwick Filter

The gyroscope predicts the rate of change of orientation, expressed as a quaternion derivative from the ω measurement and previous fused orientation as:

$$\dot{q}_{\omega,t} = 12 q_{t-1} \otimes \omega_t \quad (14)$$

The accelerometer's contribution is obtained by minimizing a gravity-alignment error function. It assumes that $\hat{a} \approx \hat{g}$. Hence, it attempts to correct the mismatch between the reference gravity vector rotated by the current orientation estimate and accelerometer reading by aligning them, as shown in eq. 15.

$$f(q_{t-1}, \hat{a}_t) = q_{t-1}^* \otimes \hat{g} \otimes q_{t-1} - \hat{a}_t \quad (15)$$

The gradient of this error with respect to the quaternion is computed using the Jacobian J , and normalized to obtain a unit correction direction:

$$\nabla f = J^T(q_{t-1}) f(q_{t-1}, \hat{a}_t), \quad \nabla f_{norm} = \frac{\nabla f}{\|\nabla f\|} \quad (16)$$

The gyroscope and accelerometer contributions are fused into a single rate of change of orientation, assuming a single step of gradient descent:

$$\dot{q}_{est,t} = \dot{q}_{\omega,t} - \beta \nabla f_{norm} \quad (17)$$

The gain β is chosen to be 0.02 in our case, so the filter relies heavily on the gyroscope for orientation propagation, while accelerometer makes minor corrections against drift at every time step. This fused rate is integrated forward and re-normalized to maintain a valid unit quaternion. Finally, the estimate is integrated with the previous orientation as:

$$\mathbf{q}_t = \hat{\mathbf{q}}_{t-1} + \dot{\mathbf{q}}_{est,t} \Delta t \quad (18)$$

IV. ANALYSIS

Each method of attitude estimation produces results with varying amounts of success. Of the four, the Madgwick filter is generally the most successful and is robust against fast perturbations as seen in Fig 3 and Fig 6. Where other methods have a high degree of error, especially with the Yaw angle, the Madgwick filter's gradient descent step correctly models all three Euler angles. The complementary filter can perform at a similar level of success to the Madgwick filter but often performs worse, likely due to errors produced from the gyroscope and accelerometer estimations which directly feed into the complementary filter. The gyroscope-only estimation often exhibits a similar shape to the Vicon baseline but has a steady offset which then affects the complementary filter's estimation ability. This steady offset increases during moments of rapid movement, illustrating the gyroscope's poor ability to handle quick movement. Conversely, the accelerometer handles rapid movement very well but can not estimate the Yaw angle. As such, the complementary filter can benefit from both sensors.

V. CONCLUSION

In this work, four estimations of attitude were examined and compared to each other and to a ground-truth baseline. Attitude estimation via only the gyroscope, only the accelerometer, via a complementary filter, and finally attitude estimation via a Madgwick filter. Through testing, the Madgwick filter was determined to be the optimal estimation of the four. However, every estimation faced some degree of error and future improvements to attitude estimation could be performed.

VI. APPENDIX

Videos can be found in this playlist: Playlist. Each video number corresponds to the .mat file index number.

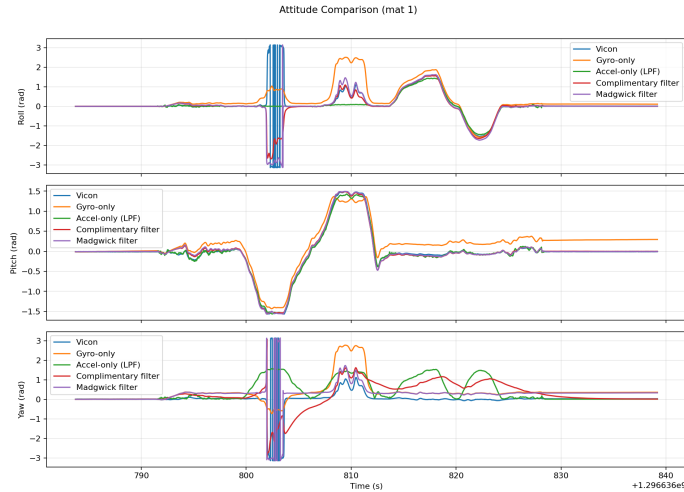


Fig. 1. Train Motion 1

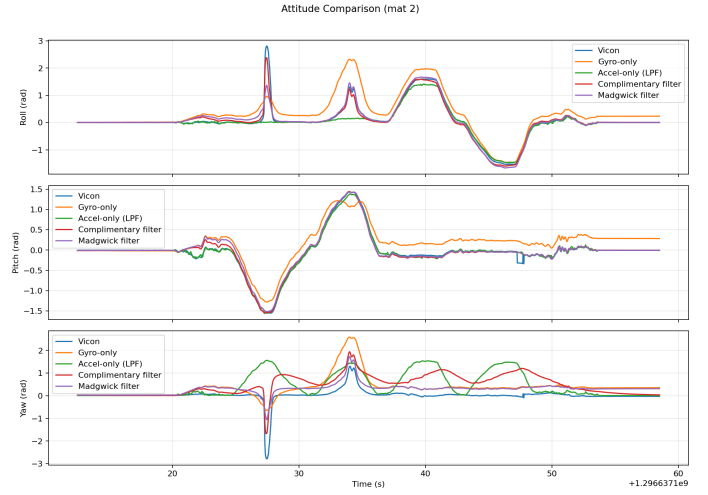


Fig. 2. Train Motion 2

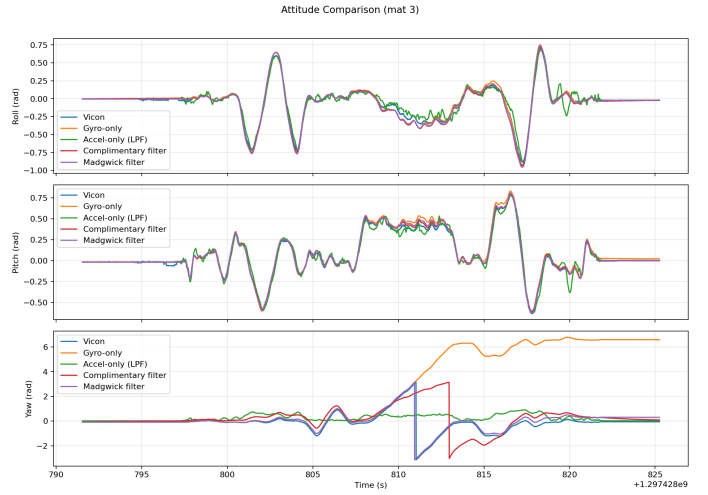


Fig. 3. Train Motion 3

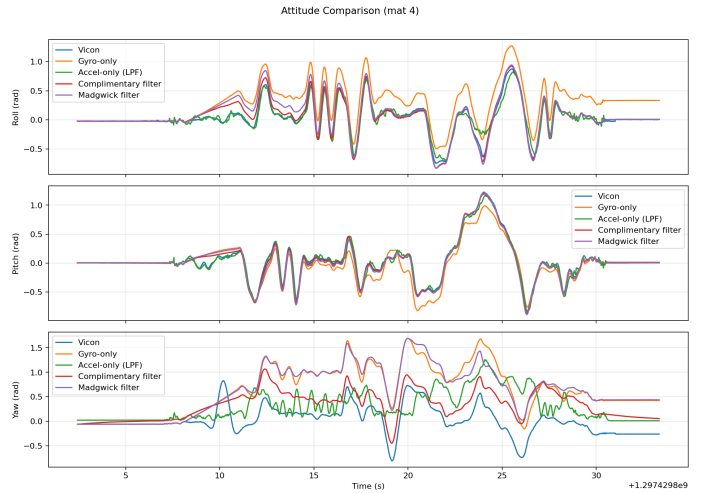


Fig. 4. Train Motion 4

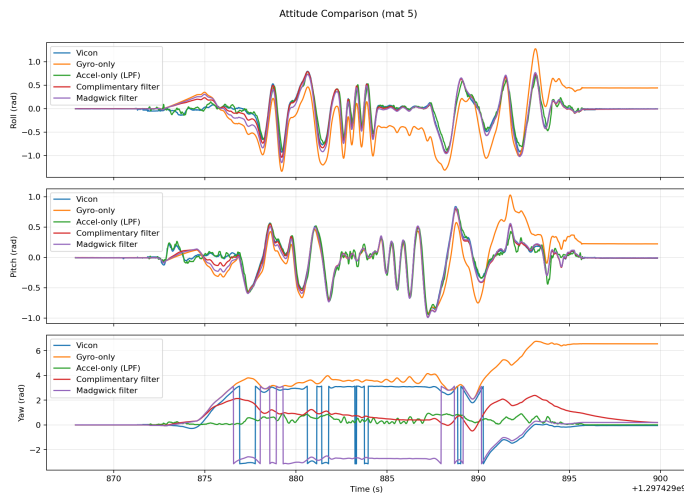


Fig. 5. Train Motion 5

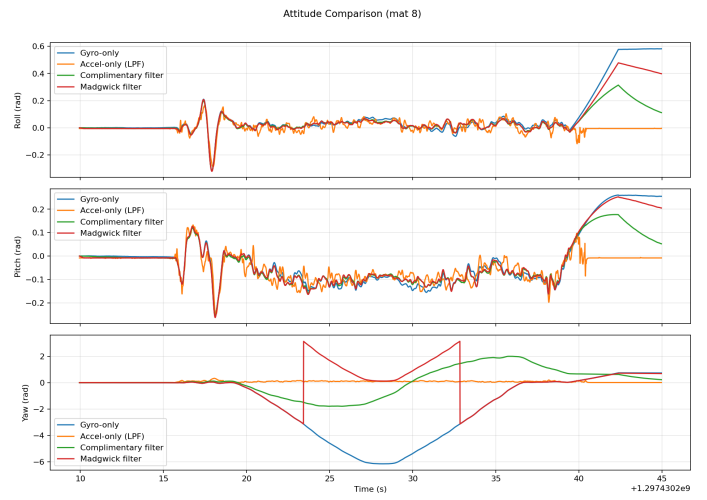


Fig. 8. Test Motion 8

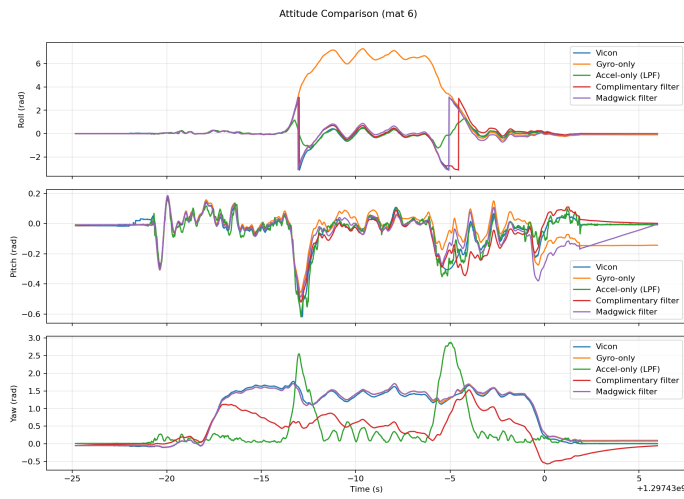


Fig. 6. Train Motion 6

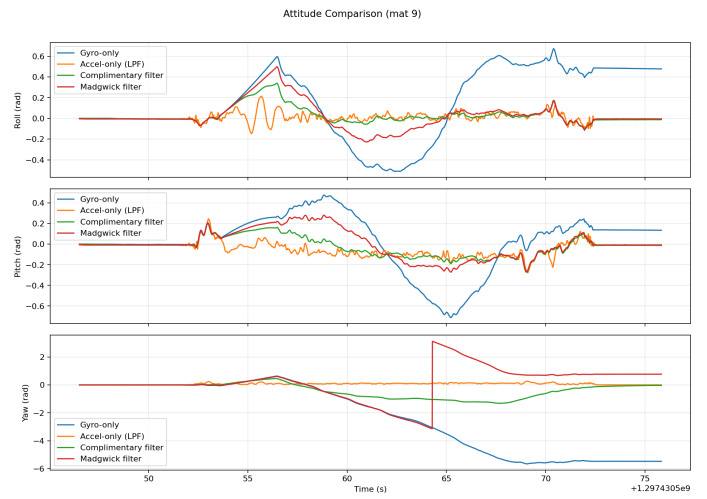


Fig. 9. Test Motion 9

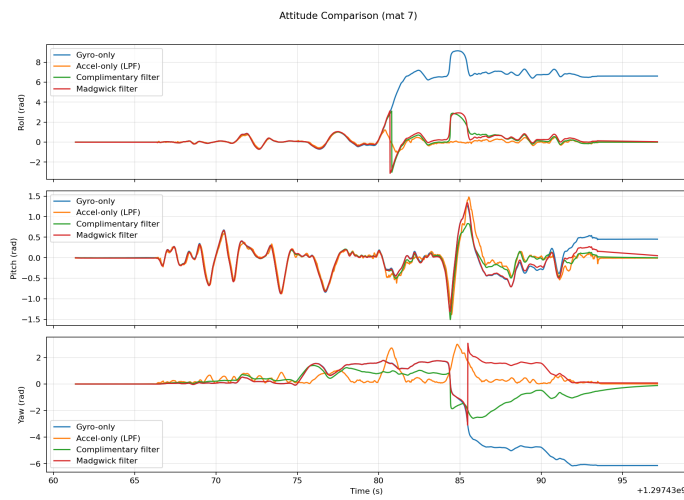


Fig. 7. Test Motion 7

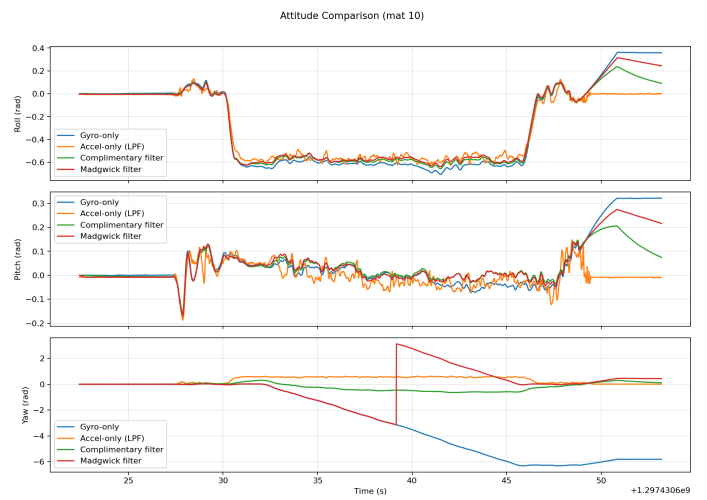


Fig. 10. Test Motion 10